

Project 1: quantifying cell movement from microscopy data

A biology lab has collected initial microscopy time courses of cells moving in 2D

The biologists want to know if treating the cells with a drug changes anything about their movement

They have collected multiple movies where they varied two experimental conditions:

- No drug added vs. treated with drug
- Cell density

A biology lab has collected initial microscopy time courses of cells moving in 2D

The lab's specific biological questions are:

- Are cells moving randomly, or do they show directional persistence?
- How fast are the cells moving on average?
- Does the drug treatment change anything about how the cells move?

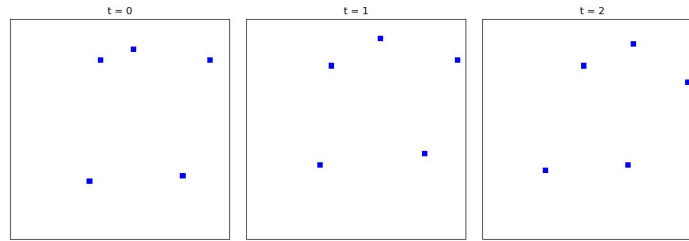
These experiments are time consuming and challenging, so they only collected an initial dataset. They want feedback before they collect more.

They bring the movies to you as their mathematical collaborator. Your goal is to identify the density range where cell movement can be estimated confidently, and to address their biological questions.

Note: in microscopy image analysis, cell segmentation can be time consuming, so we will bypass this step and work directly with cell position data

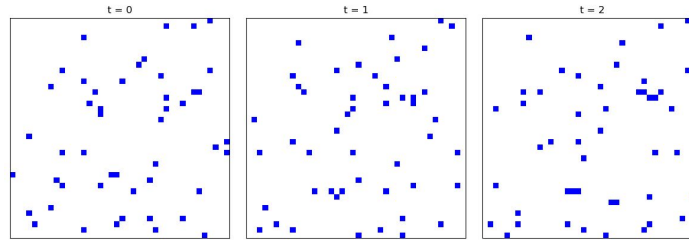
Cell centroids over time, varying density of cells:

Movie 1 (low density)



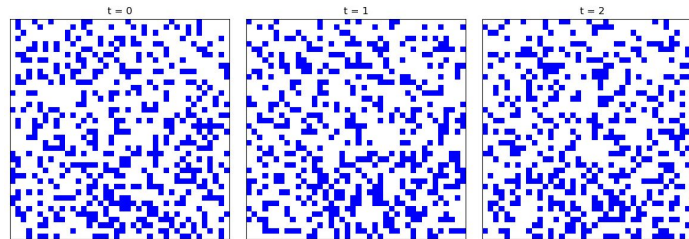
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Movie 2 (medium density)



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Movie 3 (high density)



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0 mins

5 mins

10 mins

Project 2: determining effect of an antiviral drug on cell-to-cell virus transmission

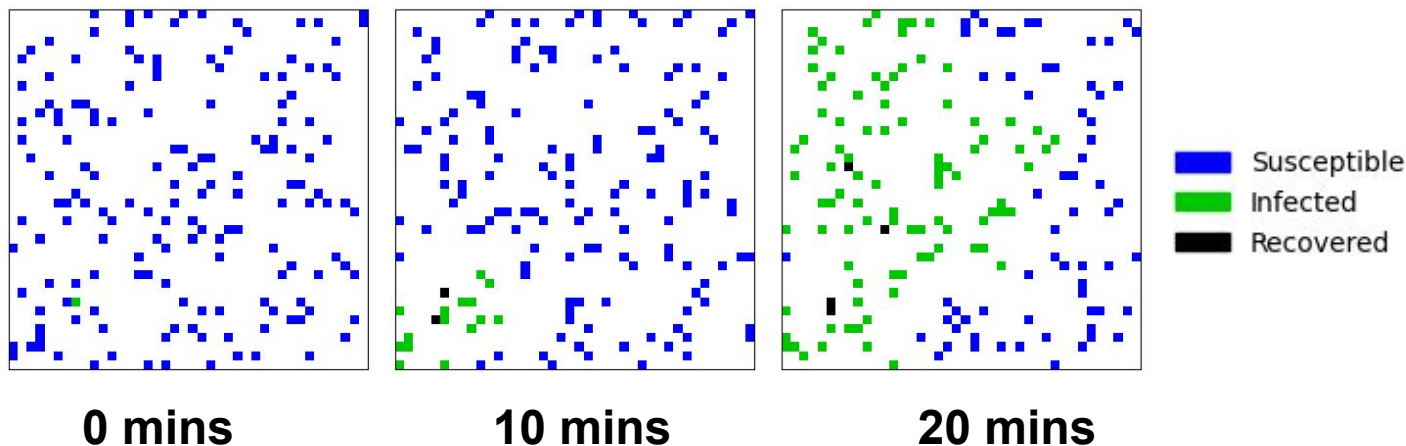
The biologist now gives you new datasets showing how a viral infection spreads over time

It has been observed that infected cells can transmit the virus to nearby susceptible cells over time.

The biologist wants to know whether adding an antiviral drug inhibits cell-to-cell transmission in some way.

Your goal is to quantify the effect that the drug has (if any) on transmission of the virus across the spatial field of cells.

Infection spread example microscopy images



Datasets begin with a small number of infected cells, and then the virus spreads cell-to-cell.

Your goal is to:

- Qualify the rate of infection
- Determine if the drug changes infection dynamics
- See what you can determine about the spatial range of infection

Taking microscopy images too frequently can expose cells to excess light and potentially affect cell behavior. The fewer images the biologists capture of each field of cells the better.

Bonus problem: For all datasets, determine the largest time interval between microscopy images that still allows the data to be analyzed reliably. This will help the biologist choose optimal imaging settings for future experiments.

Does each dataset require a different imaging time interval?

At 2:30pm tomorrow:

Your group will give a 20 minute presentation addressing the following:

- Briefly give background and introduce the problem
- Were you able to answer the biologist's questions? What approaches did you take and why?
- Based on this experience, what are some perceived challenges of working with experimentalists and how can they be overcome?
- **Document all AI prompts and plan to send to us.**
- Describe how you used generative AI:
 - How did you check whether the AI-generated suggestions were correct?
 - Which parts of the final analysis were done by your team versus assisted by AI?
 - Did you notice any limitations, mistakes, or misleading suggestions?